

Mojtaba Masoudinejad



Education

- 2021 – 2022 **PostDoc, TU Dortmund University**, Dortmund, Germany
Development of resource-aware and fault-tolerant machine learning algorithms
- 2016 – 2020 **Ph.D. (Dr.-Ing.), TU Dortmund University**, Dortmund, Germany
Modeling energy supply unit of ultra-low power devices with indoor photovoltaic harvesting
- 2008 – 2011 **M.Sc. in Automation and Robotics, TU Dortmund University**, Dortmund, Germany
Thesis: Optimization of the shuttle robots management in a high-rise warehouse
- 2002 – 2007 **B.Sc. in Electrical (Control) Engineering, K. N. Toosi University of Technology**, Tehran, Iran
Thesis: Design and implementation of Brain Emotional Learning Based Intelligent Controller for a level plant

Work experience

- 01/23 – now **Senior machine learning engineer, Bosch Center for AI (BCAI) - AI Academy**, Renningen, Germany
- Delivered diverse machine learning courses and workshops (Classical ML, Deep learning, Resource aware AI), consistently achieving higher than 4.5/5 ratings
 - Mentored and consulted on over 35 ML projects, resulting in 6 successful deployments. Projects spanned various domains, including classical and deep learning, natural language processing, and resource-aware AI
 - Developed and maintained 3 AI demonstrators, integrating them into training programs and workshops. Development includes hardware, software and machine learning components including model compression and optimization, knowledge distillation, and quantization techniques.
 - Designed and deployed a Retrieval Augmented Generation system for Training Quality Management. Including data collection, preprocessing, storage, and retrieval components, as well as the integration of the RAG system for periodic reporting and analysis of training quality metrics.
- 03/12 – 12/22 **Research associate, TU Dortmund University**, Dortmund, Germany
- Trained multiple courses and laboratories
 - Acquired and fulfilled multiple fundamental and industrial research projects
 - Organized and presented different workshops, seminars and hackathons
- 05/10 – 09/11 **Automation engineer, SABIC Polyolefine GmbH**, Gelsenkirchen, Germany
- Trained and deployed an ML-based automated system replacing manual quality control
 - Student thesis: Optimization of shuttle robots management in the warehouse
 - Student worker [full-time]: Statistical analysis of the DCS database for alarm minimization
- 07/07 – 08/08 **Automation engineer, Iran Abzar Co.**, Tehran, Iran
- Collaborated in design and purchasing of the 1st Iranian can production site
 - Head of electrical team: Designed building automation system of three Tehran metro stations
- 05/06 – 08/06 **Automation engineer, Mina Glass Co.**, Tehran, Iran
- Redesigned the batch plant automation system
- 06/05 – 09/05 **Internship, Irana Tile Co.**, Tehran, Iran
- Revised the automatic production process and provided improvement suggestions
- 07/03 – 08/03 **Electronics assembly line operator, Gam Electronics Co.**, Tehran, Iran
- Production assistant of `Kara` access control system

Skills

Programming	Python, MATLAB, bash script, C/C++, Assembly, Siemens S7
Frameworks and Packages	Scikit-learn, Keras/TensorFlow, PyTorch, PyTorch Lightning, ONNX, Optuna, MLflow, TVM, OpenCV, NumPy, pandas/polars, SciPy, matplotlib, seaborn, plotly, XGBoost, LightGBM, LibRosa, FastAPI, Jupyter Notebook, uv, Simulink/Simscape
Tools	MS Office, Git, AutoCAD
Languages	Persian (Native), English (Fluent), German (Advanced)

Honors and Awards

- 2015 Best paper award, IEEE international renewable energy congress (IREC)
- 2011 Outstanding master's thesis (top 3), TU Dortmund University
- 2011 Leading research with direct industrial impact, SABIC Polyolefine GmbH

Certificates

- Machine Learning IBM AI Engineering Specialization, Deep Learning Specialization, ML in Production, Introduction to Generative AI Learning Path Specialization
- IT skills Google IT Automation with Python Specialization
- Soft skills Rhetoric and presentation, Facilitating successful meetings, Writing successful research proposals, Knowledge management in research, First Aid Training, Fire Safety Helper Officer

Academic Activities

Positions

- 01/21 – 12/22 Postdoctoral researcher, DAES Group
TU Dortmund University , Dortmund, Germany
- 01/19 – 12/20 Researcher, Competence Center Machine Learning Rhine-Ruhr
TU Dortmund University , Dortmund, Germany
- 10/17 – 12/17 Guest researcher, Institute of Integrated Systems (IIS)
ETH Zurich , Zurich, Switzerland
- 11/16 – 12/16 Guest researcher, Institute of Integrated Systems (IIS)
ETH Zurich , Zurich, Switzerland
- 05/12 – 12/20 Research assistant, Chair of Materials Handling and Warehousing
TU Dortmund University , Dortmund, Germany

Teaching

- 01/23 – now Lecturer, Classical ML, Deep Learning, Resource-Aware AI
Bosch Center for Artificial Intelligence (BCAI) - AI Academy , Renningen, Germany
- 10/23 – now Organizer & Lecturer, One day hands-on workshop on resource-aware machine learning life-cycle
Bosch Center for Artificial Intelligence (BCAI) - AI Academy , Dortmund, Germany
- 2020 – 2021 Tutor, Computer Systems
TU Dortmund University , Dortmund, Germany
- 2020 Lecturer & organizer, Cyber-Physical Systems
TU Dortmund University , Dortmund, Germany
- 2014 – 2018 Lecturer & organizer, IT systems for Industrial Production
TU Dortmund University , Dortmund, Germany
- 2013 – 2017 Lecturer & organizer, Identification and Automation Technology
TU Dortmund University , Dortmund, Germany
- 2015 – 2017 Laboratory instructor, PLC Programming
TU Dortmund University , Dortmund, Germany
- 2013 – 2015 Laboratory instructor, Automated Identification Systems
TU Dortmund University , Dortmund, Germany

Workshops and Seminars

- 04/18 Organiser & tutor, Basics of MATLAB workshop, Chair of Materials Handling and Warehousing
TU Dortmund University , Dortmund, Germany
- 09/17 Organiser & tutor, One day hands-on workshop, Resource-aware Machine Learning Summer school
Collaborative Research Center SFB 876 , Dortmund, Germany
- 08/16 Invited talk, Ultra low power cyber-physical systems for logistics, ICT-Energy Science Conference
Aalborg University, Aalborg, Denmark , Aalborg, Denmark

Projects

- 05/19 – 12/20 Project manager, Generating a 3D annotated visual dataset for cubical objects, [The Competence Center Machine Learning Rhine-Ruhr](#)
TU Dortmund University , Dortmund, Germany
- 06/19 – 10/19 System designer, Automated bread quality control using image processing, [Fraunhofer Institute for Material Flow and Logistics](#)
[Detlef Malzers Backstube GmbH & Co. KG](#) , Dortmund, Germany
- 01/15 – 12/18 Manager, [PhyNetLab: An ultra-low power IoT testbed](#)
TU Dortmund University , Dortmund, Germany
- 04/14 – 12/18 Researcher at subproject A4, [Collaborative Research Center SFB 876](#)
TU Dortmund University , Dortmund, Germany
- 09/15 – 06/16 Researcher, FLEXINET European project
TU Dortmund University , Dortmund, Germany
- 10/15 – 12/16 Organizer, 3D printing miniature materials handling devices
TU Dortmund University , Dortmund, Germany
- 01/14 – 05/14 Developer, [COASTER: a handheld cyber-physical device for logistics applications](#)
[Fraunhofer Institute for Material Flow and Logistics](#) , Dortmund, Germany

Proposals

- 2018 DFG (German Research Foundation) project, Phase 3, [Collaborative Research Center SFB 876](#)
TU Dortmund University , Dortmund, Germany
- 2018 AiF (Alliance of Industrial Research) project, Systematic analysis of drones for safe operation
TU Dortmund University , Dortmund, Germany
- 2014 DFG (German Research Foundation) project, Phase 3, Transfer learning for human activity recognition in logistics
TU Dortmund University , Dortmund, Germany
- 2013 DFG (German Research Foundation) project, Energy optimized drive of conveyers for materials handling
TU Dortmund University , Dortmund, Germany

Publications

- [1] M. Masoudinejad, "Indoor Photovoltaic Energy Harvesting," in *Machine Learning under Resource Constraints - Applications*, 2023, pp. 195–211. doi: <https://doi.org/10.1515/9783110785982>.
- [2] M. Masoudinejad and M. Buschhoff, "PhyNetLab Test Bed," in *Machine Learning under Resource Constraints - Fundamentals*, 2023, pp. 34–45. doi: <https://doi.org/10.1515/9783110785944>.
- [3] M. Yayla, Z. Valipour Dehnoo, M. Masoudinejad, and J. Chen, "TREAM: A Tool for Evaluating Error Resilience of Tree-Based Models Using Approximate Memory," in *Embedded Computer Systems: Architectures, Modeling, and Simulation*, 2022, pp. 61–73. doi: [10.1007/978-3-031-15074-6_4](https://doi.org/10.1007/978-3-031-15074-6_4).
- [4] M. Masoudinejad, "Data-Sets for Indoor Photovoltaic Behavior in Low Lighting Conditions," *Data*, vol. 5, no. 2, p. 32, 2020, doi: [10.3390/data5020032](https://doi.org/10.3390/data5020032).
- [5] M. Masoudinejad, "Data-sets for Indoor Photovoltaic Behavior in Low Lighting condition." Zenodo, 2020. doi: [10.5281/zenodo.3697105](https://doi.org/10.5281/zenodo.3697105).
- [6] M. Masoudinejad, "Modeling Energy Supply Unit of Ultra-Low Power Devices with Indoor Photovoltaic Harvesting," phdthesis, 2020. [Online]. Available: <http://bit.ly/3U9UTig>
- [7] M. ten Hompel, M. Masoudinejad, O. Bousbib, and M. Roidl, "Automatic Identification Technology," *Operations, Logistics and Supply Chain Management*. in *Lecture Notes in Logistics*. pp. 687–718, 2019. doi: [10.1007/978-3-319-92447-2_30](https://doi.org/10.1007/978-3-319-92447-2_30).
- [8] M. Masoudinejad, M. Magno, L. Benini, and M. ten Hompel, "Average Modelling of State-of-the-Art Ultra-low Power Energy Harvesting Converter IC," in *2018 International Symposium on Power Electronics, Electrical Drives, Automation and Motion*, 2018, pp. 99–104. doi: [10.1109/SPEEDAM.2018.8445303](https://doi.org/10.1109/SPEEDAM.2018.8445303).
- [9] M. Masoudinejad, "Large Signal Converter Data from TI BQ25570." SFB876, 2018. doi: [10.17877/fd72wjj6yo](https://doi.org/10.17877/fd72wjj6yo).
- [10] M. Masoudinejad, "Large Signal Converter Data from TI BQ25505." SFB876, 2018. doi: [10.17877/fd72o6mn0g](https://doi.org/10.17877/fd72o6mn0g).
- [11] M. Masoudinejad, "Models and training datasets for TI BQ25505 and BQ25570 ICs." 2018. doi: [10.17877/FE000YSXS0](https://doi.org/10.17877/FE000YSXS0).

- [12] M. Masoudinejad, "Power analysis of PhyNode," Technical Report, 2018.
- [13] M. Masoudinejad, A. K. R. Venkatapathy, D. Tondorf, D. Heinrich, R. Falkenberg, and M. Buschhoff, "Machine Learning Based Indoor Localisation Using Environmental Data in PhyNetLab Warehouse," in *European Conference on Smart Objects, Systems and Technologies*, 2018.
- [14] R. Falkenberg et al., "PhyNetLab: An IoT-based warehouse testbed," in *Federated Conference on Computer Science and Information Systems*, 2017. doi: [10.15439/2017F267](https://doi.org/10.15439/2017F267).
- [15] M. Masoudinejad, "Automated Data Collection for Modelling Texas Instruments Ultra Low-Power Chargers," Dortmund, Technical Report, 2017. [Online]. Available: <https://bit.ly/2AtuVk9>
- [16] M. Masoudinejad, "A Power Model for DC-DC Boost Converters Operating in PFM Mode," Technical Report, 2017. [Online]. Available: <https://bit.ly/2WRnH0M>
- [17] M. Masoudinejad, A. Ramachandran Venkatapathy, J. Emmerich, and A. Riesner, "Smart Sensing Devices for Logistics Application," *Sensor Systems and Software*. 2017. doi: [10.1007/978-3-319-61563-9_4](https://doi.org/10.1007/978-3-319-61563-9_4).
- [18] M. Masoudinejad, "Photovoltaic Energy Harvesting Characteristics under Ultra-Low Light," Dortmund, Technical Report, 2016. [Online]. Available: <https://bit.ly/3cxUxe0>
- [19] S. Feldhorst, M. Masoudinejad, M. ten Hompel, and G. Fink, "Motion classification for analyzing the order picking process using mobile sensors," in *Proceedings of the 5th international conference on pattern recognition applications and methods*, 2016, pp. 706–713. doi: [10.5220/0005828407060713](https://doi.org/10.5220/0005828407060713).
- [20] M. Masoudinejad, M. Kamat, J. Emmerich, M. ten Hompel, and S. Sardesai, "A gray box modeling of a photovoltaic cell under low illumination in materials handling application," in *3rd International Renewable and Sustainable Energy Conference*, 2015, pp. 1–6. doi: [10.1109/IRSEC.2015.7455081](https://doi.org/10.1109/IRSEC.2015.7455081).
- [21] M. Masoudinejad, "A Measurement Platform for Photovoltaic Energy Harvesting in Indoor Low Light Environment," Dortmund, Technical Report, 2015.
- [22] M. Masoudinejad, J. Emmerich, D. Kossmann, A. Riesner, M. Roidl, and M. ten Hompel, "A measurement platform for photovoltaic performance analysis in environments with ultra-low energy harvesting potential," *Sustainable Cities and Society*, pp. 74–81, 2015, doi: [10.1016/j.scs.2015.09.006](https://doi.org/10.1016/j.scs.2015.09.006).
- [23] M. Masoudinejad, J. Emmerich, D. Kossmann, A. Riesner, M. Roidl, and M. ten Hompel, "Development of a measurement platform for indoor photovoltaic energy harvesting in materials handling applications," in *6th International Renewable Energy Congress*, 2015, pp. 1–6. doi: [10.1109/IREC.2015.7110938](https://doi.org/10.1109/IREC.2015.7110938).
- [24] M. Masoudinejad, S. Feldhorst, F. Javadian, and M. ten Hompel, "Reduction of Energy Consumption by Proper Speed Selection in PMSM-Driven Roller Conveyors," *IEEE Transactions on Industry Applications*, vol. 51, no. 2, 2015, doi: [10.1109/TIA.2014.2360013](https://doi.org/10.1109/TIA.2014.2360013).
- [25] M. Masoudinejad, "PhyNetLab: A Resource-Constrained Case Study for Large-Scale Logistics IoT Systems," Lisbon, 2015.
- [26] M. Roidl et al., "Performance Availability Evaluation of Smart Devices in Materials Handling Systems," in *IEEE ICC Workshops on Internet of Things*, 2014. doi: [10.1109/ICCChinaW.2014.7107857](https://doi.org/10.1109/ICCChinaW.2014.7107857).
- [27] M. Roidl, J. Emmerich, M. Masoudinejad, A. Riesner, and M. ten Hompel, "Entwicklung eines Versuchsfelds für grosse Systeme intelligenter Behälter," *Logistics Journal*, 2014, doi: [10.2195/lj_Proc_roidl_de_201411_01](https://doi.org/10.2195/lj_Proc_roidl_de_201411_01).
- [28] M. Masoudinejad, S. Feldhorst, F. Javadian, and M. ten Hompel, "Energy optimized speed regulation of permanent magnet synchronous motors for driving roller conveyors," in *Proceedings of the International Conference on Power Electronics and Drive Systems*, 2013, pp. 500–505. doi: [10.1109/PEDS.2013.6527070](https://doi.org/10.1109/PEDS.2013.6527070).
- [29] M. Masoudinejad, "Optimization of Shuttle Robots Management in a SABIC High Rise Warehouse," Master's thesis, 2011.
- [30] M. Masoudinejad, R. Khorsandi, A. Fatehi, C. Lucas, S. Fakhimi, and M. Jamali, "Real-time level plant control using improved BELBIC," in *17th IFAC World Congress*, Seoul, Korea, 2008. doi: [10.3182/20080706-5-KR-1001.3399](https://doi.org/10.3182/20080706-5-KR-1001.3399).
- [31] M. Masoudinejad and R. Khorsandi, "Design and implementation of Brain Emotional Learning Based Intelligent Controller for a level pilot plant," Bachelor's thesis, Tehran, Iran, 2007.
- [32] M. Masoudinejad and A. Beigi Khoshk, "Iran Khodro Stock Market Estimation Using Polynomial Neural Networks," in *9th Iranian Student Conference on Electrical Engineering*, 2006. [Online]. Available: <https://bit.ly/2T8c2tC>
- [33] M. Masoudinejad, "Analysis of the Karkhe Dam Automatic Voltage Regulator," in *8th Iranian Student Conference on Electrical Engineering*, Kerman, Iran, 2005. [Online]. Available: <https://bit.ly/2WwxISd>